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FACULTY OF MARITIME SCIENCE

MODELLING AND SIMULATION

Hybrid Modelling of a Fluid Flow over a Hydraulic Jump

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Abstract
Climate change, population expansion and urbanization have exposed us humans to the severe calamity of floods. Without efficient waterway design and management things could go downstream. One of the most common waterways are open channels, which flow through most cities and places of habitancy. Bedforms are common to be formed in these waterways due to turbulence, interacting with the geometry of the channel, which could result in a flood or cause a hindrance to the flow. The phenomenon that causes the discourse in flow, is known as a Hydraulic Jump. In this study, Hybrid Modelling is used to compare the results generated in a laboratory experiment for the formation of hydraulic jump using a rectangular open channel flume and compare the results obtained with the simulation results. We use the CFD solver OpenFOAM, to simulate the flow. OpenFOAM is a customized numerical solver and a pre-processing and post-processing tool incorporated with C++ libraries to process and simulate complex continuum mechanics and fluid mechanic problems.

Keywords
OpenFOAM , Hybrid Modelling, Hydraulic Jump

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Introduction

The formation of bedforms could create a hindrance to the flow of the channel causing substantial loss to life and property. Understanding the flow characteristics is extremely crucial to engineering design and safety. The development of hydraulic jumps are responsible for the generation of turbulence in the flow, which ultimately leads to varied flow characteristics through the flow.

Hydraulic Jumps occur when a fast-moving supercritical flow transitions to a more tranquil subcritical condition[7]. Hydraulic jump in open channels can be attributed to rapidly varied flow where a significant change in velocity occurs from supercritical flow to subcritical flow. This fact may owe to the presence of some structures obstructing the movement of flow in open channels. Under-shot weir or

gate is the most impressive example for hydraulic jump formation in canals where the flow undergoes high velocity under gates with upstream small depth and returns back to a higher downstream conjugate depth away from the gate with lower velocity.[8]

The hydraulic jump is characterized using the Froude Number[9]. Froude number is represented by Fr, in fluid mechanics, and in hydrology, Fr is a dimensionless quantity that is used in the indication of the influence of gravity on the motion of fluids. It is expressed as,

$$FroudeNo. = \frac{v}{\sqrt{gd}} \tag{1}$$

where,
'd' represents the depth of flow,
'g' is the acceleration due to gravity,
'v' is the celerity or gravity of the small surface wave.[9]

In this case study we use Hybrid modelling. Hybrid modelling refers to the situation where part of a model can be formulated based on the first principles and part of the model must be inferred from data because of a lack of understanding of the mechanistic details. For the case study, we use a bedridge (fig.2), and we simulate it in a channel with flow rate 0.152 $m^3 s^{-1}$, water height .335m (fig.3). Along with the simulation to generate real-time results, the experiment is conducted in a lab.

1. Methodology

Hybrid modeling is implemented to carry out the experiment. The experiment is initially carried out in the lab. The facilities at the Laboratory of Maritime Environmental Engineering at the Hochschule Emden-Leer. The laboratory is specifically designed for all questions of measuring and modeling at and on the sea. One of the main focuses is the multidimensional numerical fluid mechanics CFD. Parallel computers and all required tools for pre-processing, processing and post-processing are available in the laboratory. [2]

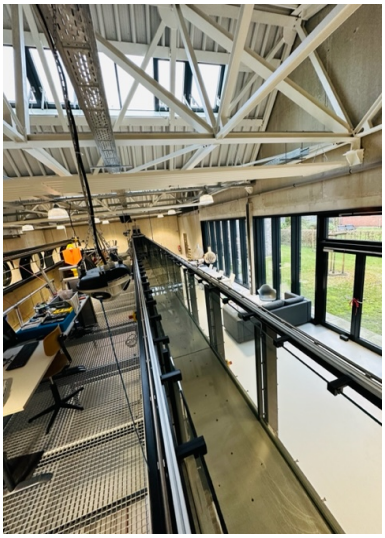


Figure 1. The Laboratory of Maritime Environmental Engineering at the Hochschule Emden-Leer

1.1 Laboratory Setup

The dimensions of the flume used for the experiment are 16 X 0.6 X 0.8.

The water height is set to 0.335m and the flow rate is set as 0.152 $m^3 s^{-1}$. A structure resembling a Bed Ridge of defined dimensions is introduced to the channel and the shape of the flow is observed. The water heights for the flow at regular intervals of distance are measured and tabulated. These results are later compared with the simulation results.

1.2 Results

The shape and the characteristics of the flow was observed, and the water height at regular intervals were measured and tabulated.

Table 1. Experimental Results

Distance in m	Water height in m
6.0	0.357
6.2	0.345
6.4	0.318
6.6	0.301
6.8	0.297
7.0	0.272
7.2	0.293
7.4	0.288
7.6	0.287
7.8	0.299
8.0	0.312
8.2	0.319
8.4	0.3045
8.6	0.323
8.8	0.357
9.0	0.319
9.2	0.311
9.4	0.353
9.6	0.321
9.8	0.337
10.0	0.342
10.2	0.323
10.4	0.350
10.6	0.335
10.8	0.327
11.0	0.340

1.3 Simulation Setup

Open Source Field Operations and Manipulation, (OpenFOAM), is the CFD solver used to conduct the experiment. OpenFOAM is a modeling toolbox based on the finite-volume method that has become widely used in the research community because of the open-source code, easy customization, and wide array of available solvers and utilities.

2. Numerical Modelling

For our model we use OpenFOAM, to simulate the flow. Our hydrodynamic solver simulated flow dynamics using the non-hydrostatic Navier–Stokes equations for single-phase flow with constant density and viscosity. The incompressible Navier-Stokes equations in the absence of body accelerations, and in two spatial dimensions, are given by;

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{2}$$

$$\frac{\partial u}{\partial t}+u(\frac{\partial u}{\partial x})+v(\frac{\partial u}{\partial y})=-(\frac{1}{\rho})(\frac{\partial p}{\partial x})+v(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}) \tag{3}$$

$$\frac{\partial v}{\partial t}+u(\frac{\partial v}{\partial x})+v(\frac{\partial v}{\partial y})=-(\frac{1}{\rho})(\frac{\partial p}{\partial y})+v(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}) \tag{4}$$

where (2) is the continuity equation and (3)-(4) are the momentum equations.[4]

Turbulence, within the flow field, was simulated using a DES approach (i.e. the Spalart-Allmaras model; Spalart and Allmaras, 1994), which employs a one-equation mixing-length RANS (Reynolds-averaged Navier–Stokes) model to approximate boundary-layer flow near walls and a large-eddy simulation (LES) scheme away from walls.

The Reynolds-averaged Navier–Stokes equations (RANS equations) are time-averaged equations of motion for fluid flow. The idea behind the equations is Reynolds decomposition, whereby an instantaneous quantity is decomposed into its time-averaged and fluctuating quantities, an idea first proposed by Osborne Reynolds. The RANS equations are primarily used to describe turbulent flows. These equations can be used with approximations based on knowledge of the properties of flow turbulence to give approximate time-averaged solutions to the Navier–Stokes equations. For a stationary flow of an incompressible Newtonian fluid, these equations can be written in Einstein notation in Cartesian coordinates as: [5]

$$\rho u_j(\frac{\partial u_j}{\partial x_j})=\rho f_i+\frac{\partial(-p\delta_i j)+\mu(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i})-\rho u_i' u_j'}{\partial x_j} \tag{5}$$

The LES scheme fully resolves turbulent flow dynamics at and above a length scale approximate to the cell dimensions of the computational grid

or mesh. Filtering procedures isolate sub-grid spatial scales, where RANS modeling assumptions are also applied to predict turbulent dynamics at that (relatively small) spatial scale.

3. Conceptual Design of the Modelling

We implement FreeCAD to design the bed ridge, an OpenSource, general purpose parametric 3D computer-aided design modeler and building information modeling software with finite element support.

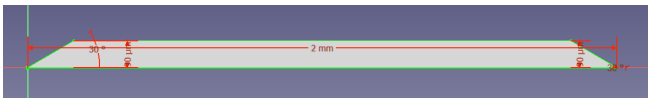


Figure 2. Bed Ridge Dimensions

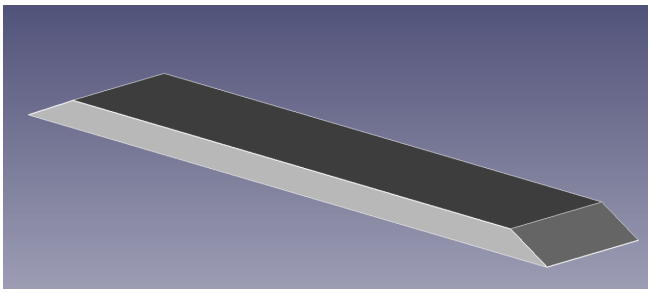


Figure 3. The Bed Ridge

3.1 Mesh Generation

Grid generation or Mesh Generation is the process of creating a mesh, or dividing a continuous geometric shape into discrete geometric or topologic cells. They are mainly used to create a discrete local approximation to the whole domain for enhanced computation. The resolution could be anywhere between 0.05 m and 0.1 m. For our experiment we chose the resolution value as (160 6 6).

To generate the grid, we access the mesh folder and access the system file blockMeshDict, to declare the vertices of the domain and create the experiment domain.

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  6                                     |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      blockMeshDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
convertToMeters 1.;

vertices
(
    (0  0   0)          // 0
    (16 0   0)          // 1
    (16 0.6 0)          // 2
    (0  0.6 0)          // 3
    (0  0   0.8)        // 4
    (16 0   0.8)        // 5
    (16 0.6 0.8)        // 6
    (0  0.6 0.8)        // 7
);

blocks
(

// insert resolution for coarse grid
    hex (0 1 2 3 4 5 6 7) (160 6 6) simpleGrading (1 1 1)  // Resolution of the coarsest grid level, at least ~10 cells per direction!!!
);

edges
(
);

patches
(
    wall xmin
    (
        (0 4 7 3)
    )
    wall xmax
    (
        (2 6 5 1)
    )
    wall ymin
    (
        (1 5 4 0)
    )
    wall ymax
    (
        (3 7 6 2)
    )
    wall zmin
    (
        (0 3 2 1)
    )
    wall zmax
    (
        (4 5 6 7)
    )
);

mergePatchPairs
(
);
// ************************************************************************* //
```

Figure 4. BlockMeshDict System File

3.1.1 Refinement Boxes

In order to get a more accurate result during simulation, the refinement boxes were configured to the shape and turbulence that was observed during the lab experiment. The refinement boxes are declared in the system file snappyHexMeshDict, within the mesh folder.

```
    format          ascii;
    class           dictionary;
    object          autoHexMeshDict;
}

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

// Which of the steps to run
castellatedMesh true;
snap            true;
addLayers       true;

// Geometry. Definition of all surfaces. All surfaces are of class
// searchableSurface.

// Surfaces are used
// - to specify refinement for any mesh cell intersecting it
// - to specify refinement for any mesh cell inside/outside/near
// - to 'snap' the mesh boundary to the surface

geometry
{

        Bed_ridge1.stl          //name of STL file
        {
            type triSurfaceMesh;
            name Bed_ridge1;
        }

// refinement boxes
        waterlevel
        {
            type searchableBox;
            min (0    0   0.260);
            max (16 0.6   0.360);
        }

        refinementbox_1
        {
            type searchableBox;
            min ( 5   0    0.120);
            max (10 0.6 0.370);
        }

        refinementbox_2
        {
            type searchableBox;
            min (5  0   0);
            max (10 0.6 0.180);
        }

//      refinementbox_3
//      {
//          type searchableBox;
//          min (7.5  0  0);
//          max (14 0.6 0.15);
//      }


    // sphere {type searchableSphere; centre (0 0 0); radius 0.1e-7;}
    // box {type searchableBox; min (20 -99 5.8); max (99 99  15.8);}
```

Figure 5. SnappyHexMeshDict System File

The refinement boxes used for the simulation are as follows:

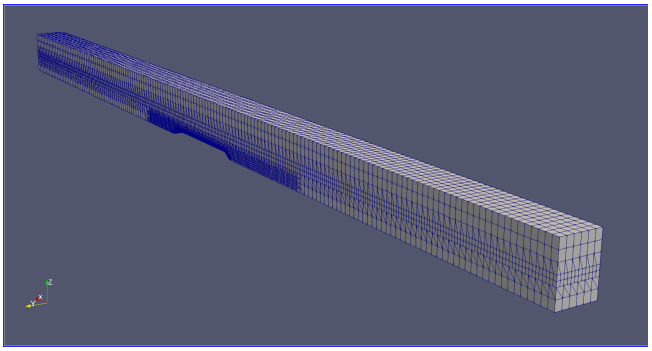


Figure 6. Mesh Generation

3.1.2 Initial and Boundary Conditions

A constant flow rate of 0.152 $m^3 s^{-1}$ is set at the inlet with a water level of 0.335 m. The water level adjusts independently to a level of 0.335 m due to the fact that the flow rate is fixed at the inlet. The body of water is defined in the setFieldsDict as shown below, which covers the whole channel.

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  5                                     |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      setFieldsDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

defaultFieldValues
(
    volScalarFieldValue alpha.water 0
    volScalarFieldValue p_rgh 1
    volVectorFieldValue U (0 0 0)
);

regions
(
    boxToCell                   // water body
    {
        box (0  0  0) (16 0.6 0.335);

        fieldValues
        (
            volScalarFieldValue alpha.water 1
            volVectorFieldValue U (0.75 0 0)
        );
    }


);

// ************************************************************************* //
~
~
~
```



```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  5                                     |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    location    "0";
    object      alpha.water;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [0 0 0 0 0 0 0];

internalField   uniform 1.;

boundaryField
{
    xmin
    {
        type            flowRateWaterAlpha;     // inlet: free water level but fixed volumetric flow rate
        refValue        0.152;
    }
    xmax
    {
        type            fixedWaterTableAlpha;   // outlet: fixed water level
        refPoint        (16 0 0.335);           // Point on water surface at the boundary
        value uniform   1.;                     // Initial value for alpha
    }
    ymin
    {
        type            zeroGradient;
    }
    ymax
    {
        type            zeroGradient;
    }
    zmin
    {
        type            zeroGradient;
    }
    zmax
    {
        type            fixedWaterTableAlpha;
        refPoint        (16 0 0.335);           // Point on water surface at the boundary
        value uniform   1.;                     // Initial value for alpha
    }
    Bed_ridge1
    {
        type zeroGradient;
    }
}


// ************************************************************************* //
~
"alpha.water" 61L, 1752B
```

Figure 8. alpha.water System File

- Calculating the velocity and declaring the velocity in the system file U. And enter the internal flow rate as 0.152m3s-1.(experimental value from the lab).



```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  8                                     |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      binary;
    class       volVectorField;
    location    "0";
    object      U;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [0 1 -1 0 0 0 0];

internalField   uniform (0.75 0 0);                 // constant velocity inside the whole domain: 0.1 m/s
                                                    // flowRateWater needs to fit to the velocity chosen
boundaryField
{
    xmin
    {
        type            flowRateWaterVelocity;      // calculates the velocity needed for fixed volumetric flow rate and given water level
        flowRateWater   0.152;                      // volumetric flow rate water [m^3/s]: 0.75m/s * 0.335m * 0.6m (velocity * water level * width)
        value           uniform (0 0 0);            // initialization boundary conditions [m/s][m/s][m/s]
    }
    xmax
    {
        type            pressureInletOutletVelocity;
        value           uniform (0 0 0);
    }
    ymin
    {
        type            slip;
    }
    ymax
    {
        type            slip;
    }
    zmin
    {
        type            slip;
    }
    zmax
    {
        type            pressureInletOutletVelocity;
        value           uniform (0 0 0);
    }
    Bed_ridge1
    {
        type            fixedValue;
        value           uniform (0 0 0);
    }
}
~
~
~
~
~
"U" [converted] 87L, 1696B
```

Figure 9. alpha.water System File

- Specific flow parameters need to be defined at the boundary patches for the system file p_rgh, and the reference points are to be defined.



```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Website:  https://openfoam.org                  |
|   \\  /    A nd           | Version:  8                                     |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      binary;
    class       volScalarField;
    location    "0";
    object      p_rgh;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [1 -1 -2 0 0 0 0];

internalField   uniform 1;

boundaryField
{
    xmin
    {
        type            zeroGradient;
    }
    xmax
    {
        type            fixedWaterTablePressure_rgh;
        mode            1;
        rho1            1.2;
        rho2            1000;
        stab1           1;
        stab2           1;
        refValue        0;
        refPoint        (16 0 0.335);
        value           uniform 0;
    }
    ymin
    {
        type            zeroGradient;
    }
    ymax
    {
        type            zeroGradient;
    }
    zmin
    {
        type            zeroGradient;
    }
    zmax
    {
        type            fixedWaterTablePressure_rgh;
        mode            1;
        rho1            1.2;
        rho2            1000;
        stab1           1;
        stab2           1;
        refValue        0;
        refPoint        (16 0 0.335);
        value           uniform 0;
    }
    Bed_ridge1
    {
        type            zeroGradient;
    }
}


// ************************************************************************* //
~
~
```

Figure 10. p.rgh System File

- Add the .stl file with the rendered 3D structure of the bed ridge to all the system files, namely, omega, k, nut, alpha.water, U.

4. Simulation

CFD solver implemented to execute the calculation is OpenFOAM. Numerical Simulations were carried out using OpenFOAM CFD code, to quantify the mean three-dimensional flow structure within a 16X0.6X0.8 m section of the channel.

The calculation is arduous, and to reduce the computational load and increase the efficiency of the solver, 4 processors are allotted to carry out the calculation. The number of processors can be allotted, modifying the DecomposeParDict system file.

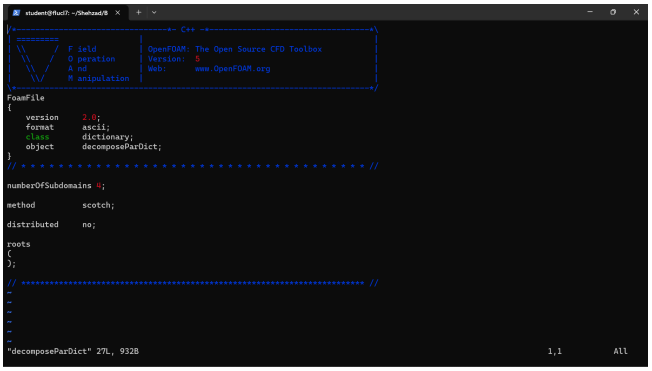


Figure 11. decomposePar System File

5. Post-Processing and Visualisation

The .foam file of the case is loaded into Paraview. The flow in the channel is visualized by choosing Outline as representation mode streamline filter.

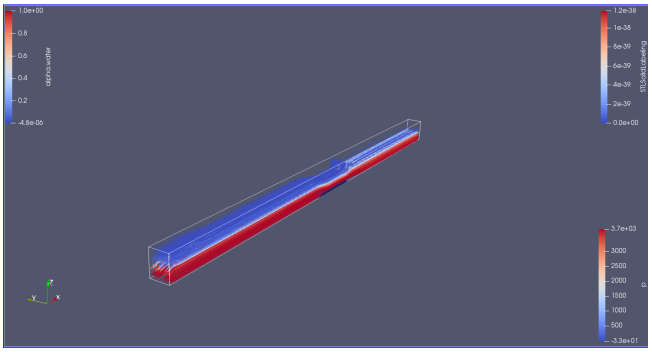


Figure 12. The Output Flow

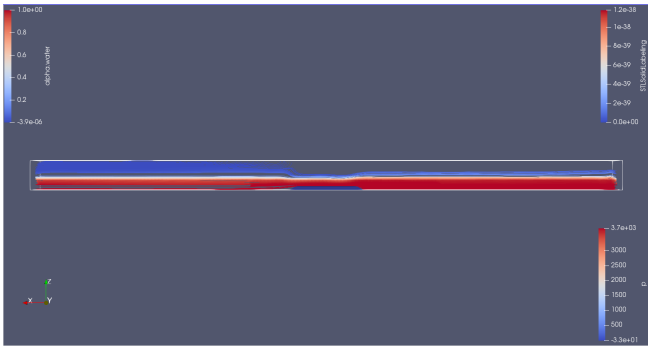


Figure 13. Sliced Cross-Sectional View of the Flow

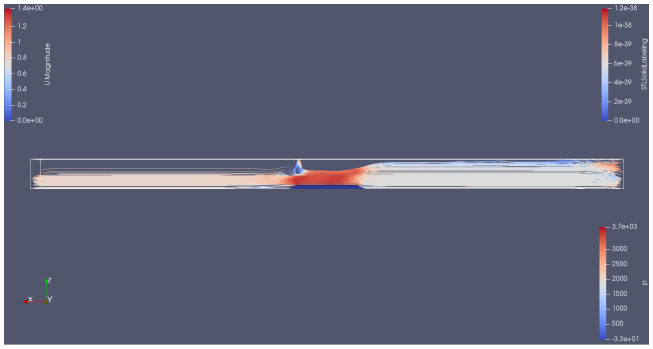


Figure 14. The Velocity Profile of the Flow at 24 Seconds

The Stream tracer filter is used to visualize the flow. The flow propagation for every 7.5 seconds of the flow is shown below.

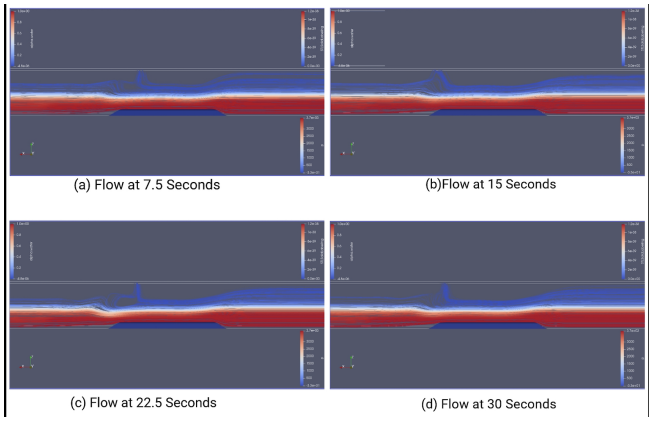


Figure 15. The Flow Propagation

5.1 Simulation Results

The Ruler tool in Paraview is used to measure the water heights of the flow at regular intervals of distance, similar to what is done in the Laboratory experiment setup.

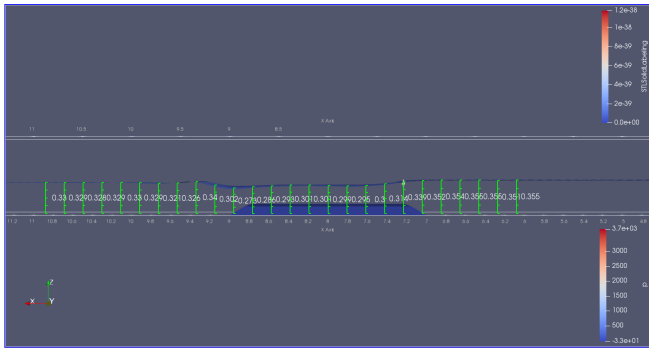


Figure 16. Simulation Results

Table 2. Simulation Results

Distance in m	Water height in m
6.0	0.355
6.2	0.351
6.4	0.355
6.6	0.355
6.8	0.354
7.0	0.352
7.2	0.339
7.4	0.314
7.6	0.300
7.8	0.295
8.0	0.299
8.2	0.301
8.4	0.3006
8.6	0.293
8.8	0.286
9.0	0.273
9.2	0.302
9.4	0.34
9.6	0.326
9.8	0.321
10.0	0.329
10.2	0.33
10.4	0.329
10.6	0.328
10.8	0.329
11.0	0.33

Table 3. Result Comparison

	Readings	
Distance in m	Experimental Results	Simulation Results
6.0	0.357	0.355
6.2	0.345	0.351
6.4	0.318	0.355
6.6	0.301	0.355
6.8	0.297	0.354
7.0	0.272	0.352
7.2	0.293	0.339
7.4	0.288	0.314
7.6	0.287	0.300
7.8	0.299	0.295
8.0	0.312	0.2299
8.2	0.319	0.301
8.4	0.3045	0.3006
8.6	0.323	0.293
8.8	0.357	0.286
9.0	0.319	0.273
9.2	0.311	0.302
9.4	0.353	0.34
9.6	0.321	0.326
9.8	0.337	0.321
10.0	0.342	0.329
10.2	0.323	0.33
10.4	0.350	0.329
10.6	0.335	0.328
10.8	0.327	0.329
11.0	0.340	0.33

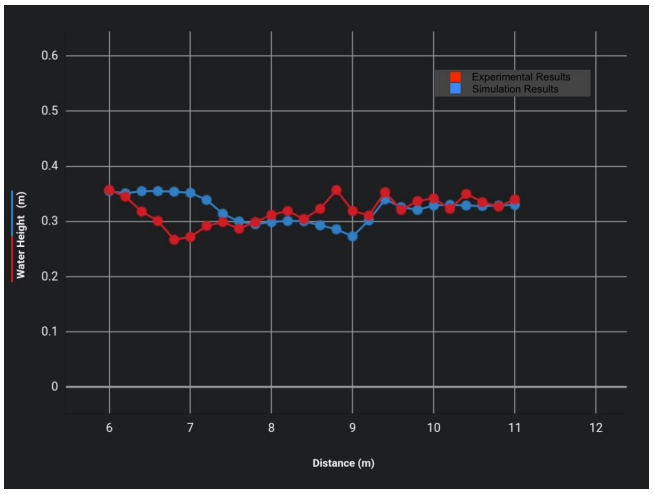


Figure 17. Result Analysis

6. Results and Discussion

The flow characteristics observed during the lab experiment and the simulation is analogous, however, there are a few notable discrepancies in the readings. The inconsistencies in the measurement could be due to the various errors caused during the experiment.

During the lab experiment the flow was extremely turbulent, resulting in inaccurate readings leading to approximate measurements. Observational errors also play a minor part in the conflicting results. This is also termed as Parallax error which occurs when the observer is unable to have a vision aligned with the pointer.[6]

7. Conclusion

A hydraulic jump occurs when there is an abrupt change in the flow velocities, leading to the turbu-

lence. Hydraulic jumps could be created due to the presence of bedforms or bed ridges or any obstruction to the flow. In the real world, this jump could mean a flood. In the study, a bed ridge was introduced in an open channel flume with a uniform flow and the flow characteristics were noted. The Hybrid model was implemented for the experiment, the results which are obtained primarily from the laboratory experiment are compared to the simulation of a flow with similar properties. Open Source Field Operation and Manipulation, OpenFOAM is the CFD solver implemented. The results were noted, and the flow characteristics and shape were observed to be similar, even though there were certain inconsistencies in the measurement due to errors. The Hybrid Modelling used, is found to be a highly effective method to simulate and study various flows.

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